Exercise 9.1

SOLUTION

1. Find remainder and quotient by simplify the following:

i. $(3x^2 - x + 2) \div (x - 1)$

Sol:

$$x-1 \;\overline{)\; 3x^2 - x + 2}$$

$$3x + 2$$

$$\pm 3x^2 \mp 3x$$

$$2x + 2$$

$$\pm 2x \mp 2$$

$$4$$

Quotient $= 3x + 2$; Remainder $= 4$

ii. $(x^3 + 12x^2 - 3x + 4) \div (x - 2)$

Sol:

$$x-2 \;\overline{)\; x^3 + 12x^2 - 3x + 4}$$

$$x^2 + 14x + 25$$

$$\pm x^3 \mp 2x^2$$

$$14x^2 - 3x + 4$$

$$\pm 14x^2 \mp 28x$$

$$25x + 4$$

$$\pm 25x \mp 50$$

$$54$$

Quotient $= x^2 + 14x + 25$; Remainder $= 54$

iii. $(x^4 - 5x^3 - 8x^2 + 13x + 12) \div (x - 6)$

Sol:

$$x-6 \;\overline{)\; x^4 - 5x^3 - 8x^2 + 13x + 12}$$

$$x^3 + x^2 - 2x + 1$$

$$\pm x^4 \mp 6x^3$$

$$x^3 - 8x^2 + 13x + 12$$

$$\pm x^3 \mp 6x^2$$

$$-2x^2 + 13x + 12$$

$$\mp 2x^2 \pm 12x$$

$$x + 12$$

$$\pm x \mp 6$$

$$18$$

Quotient $= x^3 + x^2 - 2x + 1$; Remainder $= 18$

iv. $(5x^4 - 3x^3 + 2x^2 - 1) \div (x^2 + 4)$

Sol:

$$x^2+4 \;\overline{)\; 5x^4 - 3x^3 + 2x^2 - 1}$$

$$5x^2 - 3x - 18$$

$$\pm 5x^4 \qquad \pm 20x^2$$

$$-3x^3 - 18x^2 - 1$$

$$\mp 3x^3 \qquad \mp 12x$$

$$-18x^2 + 12x - 1$$

$$\mp 18x^2 \qquad \mp 72$$

$$12x + 71$$

Quoti

ent $= 5x^2 - 3x - 18$; Remainder $= 12x + 71$

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v. $\left(3x^4-5x^3+4x-6\right)\div\left(x^2-3x+5\right)$

Sol:

$$\begin{array}{r|l} & 3x^2+4x-3 \\ \hline x^2-3x+5 & 3x^4-5x^3+4x-6 \\ & \pm 3x^4 \mp 9x^3 \pm 15x^2 \\ \hline & \quad 4x^3-15x^2+4x-6 \\ & \quad \pm 4x^3 \mp 12x^2 \pm 20x \\ \hline & \quad\quad -3x^2-16x-6 \\ & \quad\quad \mp 3x^2 \pm 9x \mp 15 \\ \hline & \quad\quad\quad -25x+9 \end{array}$$

Quotient $=3x^2+4x-3$; Remainder $=-25x+9$

2. Use the remainder theorem to find the remainder when the first polynomial is divided by the second polynomial.

i) $x^2+5x+6\ ,\ x-2$

Sol: Let $f(x)=x^2+5x+6$ & $x-a=x-2\Rightarrow a=2$

By remainder theorem:

Remainder $=f(2)=(2)^2+5(2)+6$

$=4+10+6=20$

نوٹ:

Remainder معلوم کرنے کے لیے دیے گئے function کو f(x) کے برابر کر کے اس میں x کی قیمت put کریں گے تو remainder آجائے گا۔

ii) $x^3+5x^2+6\ ,\ x+1$

Sol: Let $f(x)=x^3+5x^2+6$

$x-a=x+1\Rightarrow a=-1$

By using remainder theorem:

Remainder $=f(-1)=(-1)^3+5(-1)^2+6$

$=-1+5+6=10$

iii) $x^4+x^3+x^2+x+1\ ,\ x-1$

Sol: Let $f(x)=x^4+x^3+x^2+x+1$

$x-a=x-1\Rightarrow a=1$

By using remainder theorem:

Remainder $=f(1)=(1)^4+(1)^3+(1)^2+(1)+1=5$

iv) $x^4+x^2+1\ ,\ x+3$

Sol: Let $f(x)=x^4+x^2+1$

$x-a=x+3\Rightarrow a=-3$

By using remainder theorem:

Remainder $=f(-3)=(-3)^4+(-3)^2+1$

$=81+9+1=91$

v) $x^4+x^3+2\ ,\ x+2$

Sol: Let $f(x)=x^4+x^3+2$

$x-a=x+2\Rightarrow a=-2$

By using remainder theorem:

Remainder $=f(-2)=(-2)^4+(-2)^3+2$

$=16-8+2=10$

3. Use the factor theorem to determine if the fi polynomial is a factor of the second polynomial.

i) $x+1\ ,\ x^2-1$

Sol: Here $f(x)=x^2-1$ and $a=-1$

$f(-1)=(-1)^2-1=1-1=0$

So, $x+1$ is factor of x^2-1

نوٹ:

یہ چیک کرنے کے لیے کہ دیا گیا polynomial اس کا factor ہے یا نہیں صرف remainder معلوم کریں اگر remainder=0 ہو تو factor ہے اور اگر remainder $\neq 0$ ہو تو factor نہیں ہے۔

ii) $x-2\ ,\ x^2-5x+6$

Sol: Here $f(x)=x^2-5x+6$ and $a=2$

$f(2)=(2)^2-5(2)+6$

$f(2)=4-10+6=0$

So, $x-2$ is factor of x^2-5x+6

iii) $x+1\ ,\ x^3+x^2+x-3$

Sol: Here $f(x)=x^3+x^2+x-3$ and $a=-1$

$f(-1)=(-1)^3+(-1)^2+(-1)-3$

$f(-1)=-1+1-1-3=-4\neq 0$

So, $x+1$ is not factor of x^3+x^2+x-3

iv) $x-2\ ,\ x^3+x^2-7x+2$

Sol: Here $f(x)=x^3+x^2-7x+2$ and $a=2$

$f(2)=(2)^3+(2)^2-7(2)+2$

$f(2)=8+4-14+2=0$

So, $x-2$ is factor of x^3+x^2-7x+2

v) $x-3\ ,\ x^4-3x^3+x^2-x+1$

Here $f(x)=x^4-3x^3+x^2-x+1$ and $a=3$

$f(3)=(3)^4-3(3)^3+(3)^2-3+1$

$f(3)=81-81+9-3+1=7\neq 0$

So, $x-3$ is not factor of $x^4-3x^3+x^2-x+1$

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4. **Use synthetic division to show that x is the zero of the polynomial and use the result to factorize the polynomial completely.**

i) $x^3 - 7x + 6 \ , \ x = 2$

Sol: Let $f(x) = x^3 - 7x + 6$

$= x^3 + 0x^2 - 7x + 6$

Here $a = 2$

2	1	0	-7	6
		2	4	-6
	1	2	-3	0

$Q(x) = x^2 + 2x - 3$

$= x^2 + 3x - x - 3 = x(x+3) - 1(x+3)$

$= (x-1)(x+3)$

Therefore, other two factors are $(x-1)$ and $(x+3)$

Hence $x^3 - 7x + 6 = (x-2)(x-1)(x+3)$

> نوٹ:
> ان سوالات میں جب $f(x)$ لکھتا ہے تو اگر x کی کوئی power نہیں دی گئی تو اس کی جگہ 0 لگاتا ہے۔

ii) $x^3 - 28x - 48 \ , \ x = -4$

Sol: Let $f(x) = x^3 - 28x - 48$

$= x^3 + 0x^2 - 28x - 48$

Here $a = -4$

-4	1	0	-28	-48
		-4	16	48
	1	-4	-12	0

$Q(x) = x^2 - 4x - 12$

$= x^2 - 6x + 2x - 12$

$= x(x-6) + 2(x-6)$

$= (x+2)(x-6)$

So, other two factors are $(x+2)$ and $(x-6)$

Hence $x^3 - 28x - 48 = (x+4)(x+2)(x-6)$

iii) $2x^4 + 7x^3 - 4x^2 - 27x - 18 \ , \ x = 2 \ , \ x = -3$

Sol: Let $f(x) = 2x^4 + 7x^3 - 4x^2 - 27x - 18$

Step 1: use synthetic division with $x = 2$ or $a = 2$

2	2	7	-4	-27	-18
		4	22	36	18
	2	11	18	9	0

As remainder = 0 so, $(x-2)$ is factor of new polynomial $2x^3 + 11x^2 + 18x + 9$.

Step 2: use synthetic division $x = -3$ or $a = -3$

Again

-3	2	11	18	9
		-6	-15	-9
	1	2	-3	0

Remainder = 0 , so $(x+3)$ is also factor of new polynomial.

$2x^2 + 5x + 3 = 0 \Rightarrow 2x^2 + 2x + 3x + 3 = 0$

$2x(x+1) + 3(x+1) = 0 \Rightarrow (2x+3)(x+1) = 0$

Other two factors are $(2x+3)$ and $(x+1)$

So, complete factors of

$2x^4 + 7x^3 - 4x^2 - 27x - 18$

$= (x-2)(x+3)(x+1)(2x+3)$

Q5. **Use synthetic division to find the quotient and the remainder when the polynomial $x^4 - 10x^2 - 2x + 4$ is divided by $x + 3$.**

Sol:

-3	1	0	-10	-2	4
		-3	9	3	-3
	1	-3	-1	1	1

Here $f(x) = x^4 + 0x^3 - 10x^2 - 2x + 4$ and $a = -3$

So, when $x^4 - 10x^2 - 2x + 4$ is divided by $x + 3$ using synthetic division.

Quotient $= x^3 - 3x^2 - x + 1$ and remainder = 1

6. **If $x + 1$ and $x - 2$ are factors of $x^3 - px^2 + qx + 2$ Using synthetic division, find the values of p and q.**

Sol: As given that $x + 1$ and $x - 2$ are factors of the polynomial $f(x) = x^3 - px^2 + qx + 2$

This means: $f(-1) = 0$ and $f(2) = 0$

By using synthetic division

-1	1	-p	q	2
		-1	P+1	-p-q-1
	1	-p-1	p+q+1	-p-q+1

As $x + 1$ is factor so, $-p - q + 1 = 0 ----(a)$

2	1	-p	q	2
		2	-2p+4	-4p+2q+8
	1	-p+2	-2p+q+4	-4p+2q+10

As $x - 2$ is factor so, $-4p + 2q + 10 = 0 ----(b)$

By $(b) - 4(a)$ we get

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$-\cancel{4p}+2q+10=0$

$\mp\cancel{4p}\mp 4q\pm 4=0$

$6q+6=0 \Rightarrow 6q=-6$

$\Rightarrow q=\frac{\cancel{-6}^{-1}}{\cancel{6}} \Rightarrow q=-1$

By using $q=-1$ in (a) we get

$-p-(-1)+1=0$

$-p+1+1=0$

$-p+2=0$

$\cancel{-}p=\cancel{-}2 \Rightarrow p=2$

So, we have $p=2$ and $q=-1$

Q7. **When the polynomial $4x^4+2x^3+kx^2+13$ is divided by $x+1$, the remainder is 16. Find the value of k.**

Sol: Let $f(x)=4x^4+2x^3+kx^2+13$

Substitute $x=-1$

$f(-1)=4(-1)^4+2(-1)^3+k(-1)^2+13$

Put $f(-1)=16$

$16=k+15$

$16-15=k \Rightarrow 1=k$ or $k=1$

Q8. **When the polynomial x^3+x^2+x+k is divided by $x+1$, the remainder is 7. Find the value of k.**

Sol: Let $f(x)=x^3+x^2+x+k$

Substitute $x=1$

$f(1)=(1)^3+(1)^2+(1)+k$

Put $f(1)=7$

$7=3+k$

$7-3=k \Rightarrow k=4$

9. **Use factor theorem to find the values of p and q if $x+1$ and $x-2$ are the factors of the polynomial x^3+px^2+qx+3.**

Sol: By using factor theorem if $x+1$ is a factor then $f(-1)=0$ & if $x-2$ is a factor then $f(2)=0$

So, $f(-1)=(-1)^3+p(-1)^2+q(-1)+3$

$f(-1)=-1+p-q+3$

Put $f(-1)=0$

$\Rightarrow p-q+2=0----(a)$

So, $f(2)=(2)^3+p(2)^2+q(2)+3$

$f(2)=8+4p+2q+3$

Put $f(2)=0$

$4p+2q+11=0----(b)$

From eq (a) $p=q-2$

Substitute into eq (b)

$4(q-2)+2q+11=0$

$4q-8+2q+11=0$

$6q+3=0$

$6q=-3$

$q=\frac{-\cancel{3}^{1}}{\cancel{6}^{2}} \Rightarrow q=\frac{-1}{2}$

By using $p=q-2$

$p=\frac{-1}{2}-2 \Rightarrow \frac{-1-4}{2}=\frac{-5}{2}$

So, $p=\frac{-5}{2}$

Q10. **Use factor theorem to find the values of a & b if -2 & 2 are the roots of the polynomial $2x^3+4x^2+ax+b$.**

Sol: Let $f(x)=2x^3+4x^2+ax+b$

$f(2)=2(2)^3+4(2)^2+a(2)+b$

$f(2)=2(8)+4(4)+2a+b$

$f(2)=16+16+2a+b$

$f(2)=32+2a+b$

Given 2 is a root so, $f(2)=0$

$\Rightarrow 32+2a+b=0----(a)$

Now $f(-2)=2(-2)^3+4(-2)^2+a(-2)+b$

$f(-2)=2(-8)+4(4)-2a+b$

$f(-2)=-\cancel{16}+\cancel{16}-2a+b$

$f(-2)=-2a+b$

Given -2 is a root so, $f(-2)=0$

$\Rightarrow -2a+b=0-----(b)$

by adding eq. (a) & (b)

we have

$-\cancel{2a}+b=0$

$\cancel{2a}+b+32=0$

$2b+32=0 \Rightarrow 2b=-32$

$b=\frac{\cancel{-32}^{-16}}{\cancel{2}} \Rightarrow b=-16$

By using eq. (b)

$-2a+b=0$

$-2a+(-16)=0$

$-2a=16 \Rightarrow a=\frac{\cancel{-16}^{-8}}{\cancel{2}} \Rightarrow a=-8$

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